

INNOVUS

Physical Design Command Reference

Categories Covered

PD Flow: Floorplan | Power Plan | Placement | CTS | Routing
Selection: Instance | Net | Pin | Object | InstByCellName
Highlight | GUI / View Controls | Timing & DRV | Scan Chain
dbGet / dbSet Queries | Design Status | ECO & Optimization

Source: Cadence Innovus TCR v23.11 | dbGet Commands Guide

1. FLOORPLAN COMMANDS

Command / Syntax	Description / Notes
floorPlan	Create/initialize the floorplan with core area, IO ring, etc.
dbGet top.fplan.coreBox	Get core area bounding box coordinates
dbGet top.fplan.coresite.size	Get core site dimensions
dbGet top.fplan.ioBox	Get IO boundary coordinates
dbGet top.fplan.ios	List all IO pads in the design
dbGet top.fplan.core2left	Get distance between core and left pad
dbGet top.fplan.rows	Get all row objects
dbGet [dbGet top.fplan.rows].box	Get coordinates of all placement rows
dbGet top.fplan.rBlkgs.shapes.rect	Get points of rectangular routing blockages
dbGet top.fplan.rBlkgs.shapes.poly	Get points of rectilinear routing blockages
dbGet top.fplan.pBlkge.type	Get placement blockage type (soft / hard / partial)
dbGet top.fplan.pblkge.bboxes	Get blockage coordinates (without cell view)
dbGet top.fplan.pblkge.box	Get blockage coordinates (with cell view)
dbGet top.fplan.pblkgs.type hard	Filter hard placement blockages
dbGet [dbGet -p top.fplan.pBlkgs type hard].boxes	Coordinates of hard placement blockages
createPlaceBlockage -type hard -box {x1 y1 x2 y2}	Create a hard placement blockage
deletePlaceBlockage	Delete a placement blockage
deleteRouteBlk	Delete a routing blockage
deleteAllFPObjects	Delete all floorplan objects
loadFPlan <file>	Load floorplan from file
saveFPlan <file>	Save floorplan to file
saveIoFile <file>	Save IO pin assignment file
loadIoFile <file>	Load IO pin assignment file
finishFloorplan	Finalize floorplan after all FP settings
dbShape -output polygon [dbGet top.fPlan.bboxes]	Report polygon points forming the die area
initCoreRow	Initialize core rows
cutRow	Cut a placement row (used to create row gaps)
deleteRow	Delete a row from the floorplan
setObjFPlanBox Instance	Place/resize an instance using FP box coordinates

Command / Syntax	Description / Notes
<Inst_name> <coords>	
setObjFPlanBoxList	Set multiple instance FP boxes in one call
setObjFPlanPolygon	Set instance boundary using a polygon
queryFPlanObject	Query a floorplan object at a given coordinate
resizeFloorplan	Resize the floorplan boundary
fix_floorplan	Fix overlapping or misaligned FP objects
fix_boundary_overlaps	Fix boundary overlap issues
flipModule	Flip a hierarchical module
flipOrRotateObject	Flip or rotate selected FP object
move_obj	Move a floorplan object to a new location
alignObject	Align selected objects in the floorplan
snapFPlan	Snap floorplan objects to grid
snapFPlanIO	Snap IO objects to grid
spaceIOInst	Space IO instances evenly
placeAIO	Place analog IO instances
placePadIO	Place pad IO instances
placePIO	Place physical IO instances
fixAllIos	Fix all IO pin placements
legalizeFPlan	Legalize the floorplan (remove overlaps)
predict_floorplan	Predict optimal floorplan dimensions based on design
generate_fence	Generate a fence region for hierarchical partitioning
get_macro_place_constraint	Get macro placement constraint settings
set_macro_place_constraint	Set macro placement constraints
refine_macro_place	Refine macro placement
refineMacro	Refine individual macro placement
report_narrow_channel	Report narrow routing channels between macros
dbGet [dbGet -p2 top.insts.cell.baseclass block].name	Get names of all macro instances
dbGet [dbGet -p top.insts.cell.baseclass block].name	Get names of macro cells
dbGet [dbGet -p2 top.insts.cell.baseclass block].orient	Get orientation of all macros
dbGet [dbGet -p2 top.insts.cell.baseclass block].box	Get bounding box coordinates of all macros
dbGet [dbGet -p2 top.insts.cell.baseclass core].name	Get names of all standard cells

Command / Syntax	Description / Notes
dbGet [dbGet -p2 top.insts.cell.baseclass pad].name	Get names of all IO pad instances
dbGet head.dbUnits	Get database unit setting (user units per micron)
dbGet head.mfgGrid	Get manufacturing grid value

2. POWER PLANNING COMMANDS

Command / Syntax	Description / Notes
addRing	Add power/ground ring around core or macro
addStripe	Add power/ground stripes across the design
sroute	Route special (PG) nets including follows-pin, stripes, rings
editPowerVia	Add or edit power vias between metal layers
deleteAllPowerPrerouts	Delete all pre-routed power/ground wires (rings, stripes)
deleteAllSignalPreroutes	Delete all pre-routed signal wires
editDelete -shapes {RING STRIPE} -status {ROUTED FIXED}	Delete RING/STRIPE shapes with status ROUTED or FIXED
editDelete -type Signal - status {ROUTED FIXED}	Delete FIXED/ROUTED signal nets
addTieHilo -cell TIELO -prefix TIE	Add tie-hi / tie-lo cells for constant connections
dbGet top.insts.instTerms.isTieHi 1 dbGet top.insts.instTerms.isTieLo 1	Check for tiehi/lo connections not yet assigned a tie cell. <i>Returns '0x0' if all OK; returns '1's if tie cells are missing.</i>
dbGet [dbGet -p top.insts.instTerms.isTieHi 1].name dbGet [dbGet -p top.insts.instTerms.isTieLo 1].name	List the instTerms still needing a tiehi / tielo cell
dbGet [dbGet -p2 top.insts.cell.subClass coreTieLo].instTerms.net.allTe rms.isInput 1].name	Get all insTerm names tied to tieLo cells
dbGet [dbGet -p head.libCells.subClass coreTieHigh].name dbGet [dbGet -p head.libCells.subClass coreTieLow].name	Get library cell names for tie-high / tie-low cells
addNetnewName	Add a new net to the design by name
AttachTerm <Inst_name> <Pin_name> <net_name>	Attach an instance terminal to a net
deletePGPin	Delete a power/ground pin
set_pg_nets	Define power/ground net associations

Command / Syntax	Description / Notes
checkNetlist	Check netlist for floating inputs, multi-driven nets, PG connections
checkDesign -all	Comprehensive design rule/integrity check

3. PLACEMENT COMMANDS

Command / Syntax	Description / Notes
place_design	Run global placement (full flow command)
placeInstance inst_name x y	Place a single instance at specified coordinates
AddInst -cell <cell_name> -inst <Inst_name> -loc X1 Y1	Add and place a new instance
refinePlace	Refine placement after global placement
setPlaceMode -congEffort medium	Set congestion effort level (low/medium/high)
setPlaceMode -modulePadding <inst_name> <factor>	Set module padding factor for an instance
queryPlaceDensity	Query placement density across the design
QueryDensityInBox x1 y1 x2 y2	Query placement density in a specific area
CreateDensityArea x1 y1 x2 y2 50%	Create a density area constraint (e.g. 50% target)
trialRoute -maxRouteLayer 6	Run trial routing up to specified metal layer
specifyCellPad <cell_name> <factor>	Specify placement padding around a specific cell
setInstancePlacementStatus	Set instance placement status (fixed/placed/unplaced)
dbSet [dbGet -p top.insts.name *clk*].pStatus fixed	Fix all instances whose name matches *clk*
dbGet [dbGet -p top.insts.pStatus unplaced].name	Get list of unplaced instances
dbGet [dbGet -p top.insts.pStatus placed].name	Get list of placed instances
dbGet [dbGet -p top.insts.pStatus fixed].name	Get list of fixed instances
dbGet [dbGetInstByName instName].pStatus	Get placement status of a specific instance
dbGet -u top.insts.cell.name	Get list of all unique cell types in the design
dbGet top.insts.cell.name BUF*	Get all buffer cell instances
dbGet top.insts.cell.name *FF*	Get all flip-flop cell instances
dbGet [dbGet -p2 top.insts.cell.isSequential 1].name	Get all sequential (register) instances
all_registers all_registers -flops	Get all registers; optionally only flops (Timing Tcl)

Command / Syntax	Description / Notes
sizeof_collection [all_registers -flops]	Count total number of flip-flops
setEndcapMode -help	See all endcap mode options
setEndCapMode -leftTopCorner FILL4 -leftTopEdge FILL4 -leftBottomEdge FILL4 - leftBottomCorner FILL4 -prefix end_left setEndCapMode -rightEdge FILL2 -leftEdge FILL2 -topEdge {FILL2 FILL32} - bottomEdge {FILL2 FILL32}	Configure endcap cells for all corner/edge positions
addEndCap -preCap <filler_name> -postCap <filler_name> -prefix name	Add endcap cells to row ends
addEndCap	Add endcap cells using current setEndCapMode settings
addWellTap -cell FILL2 - cellInterval 20 -checkerBoard	Add well-tap cells at 20-unit intervals in checkerboard pattern
deleteFiller -inst <instname> deleteFiller -cell FILL2	Delete specific filler instances or all fillers of a cell type
dbGet -regexp [dbGet -p2 top.insts.cell.baseClass core].name WELLTAP	Find all well-tap instance names via regexp
dbGet [dbGet -p top.insts.isPhysOnly 1].name	Get names of all physical-only cells (filler, endcap, etc.)
dbGet [dbGet -p head.libCells.subClass coreWellTap].name	Get well-tap cell names from library
dbGet [dbGet -p head.libCells.subclass coreEndCap*].name	Get endcap cell names from library
dbGet [dbGet -p head.libCells.subClass coreSpacer].name	Get spacer/filler cell names from library
dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloTop dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloBot dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloLeft dbGet [dbGet -p2 top.insts.cell.subClass block*].pHaloRight	Get placement halo sizes (top/bot/left/right) for all blocks
dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloSideSize dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloBotLayer.name dbGet [dbGet -p2 top.insts.cell.subClass block*].rHaloTopLayer.name	Get routing halo size and metal layer range for all blocks
dbGet [dbGet -p2 top.insts.cell.name cellName].name	Get all instances of a specific cell type

Command / Syntax	Description / Notes
dbGet [dbGetCellByName cellName].size	Get cell size from the library
dbGet [dbGetCellByName cellName].baseClass dbGet [dbGetCellByName cellName].subClass	Get base class and sub-class of a library cell
attachIOBuffer -basename new1 -out BUFX3	Attach IO buffer to an IO pin
optDesign -preCTS [-incr]	Run pre-CTS optimization (optionally incremental)
reportGateCount	Report gate count summary
all_inputs all_outputs all_clocks	Tcl collections: all input/output ports, all clocks
sizeof_collections [all_outputs]	Count all output ports
dbGet top.inst.instTerms.net.name	Check for floating inputs (check numofOutput)

4. CLOCK TREE SYNTHESIS (CTS) COMMANDS

Command / Syntax	Description / Notes
createClockTreeSpec -file name.ctstch -bufferlist {CLKIN VX2 CLKIN VX4 ...}	Create clock tree specification file with buffer list
specifyClockTree -file name.ctstch	Specify the clock tree from a spec file
clockDesign -specfilename name.ctstch	Run clock tree synthesis using a spec file
get_clocks	Get all clock objects in the design
get_property [all_clocks] period	Get period of all clocks
get_property [all_clocks] sources	Get source pins of all clocks
get_property [get_clocks] clock_network_pins	Get clock network (sink) pins
changeClockStatus - nofixedBuffers -all	Change clock status: remove fixed flag from all buffers
deleteClockTree -all	Delete the entire clock tree
cleanUpSpecifyClockTree	Clean up clock tree specification
dbGet -regexp [dbGet top.nets].name clk	Get clock net names by regexp pattern
dbGet [dbGet -p top.nets.isCTSclock 1].name	Get net names marked as CTS clocks
dbGet [dbGet -p top.nets.isClock 1].name	Get nets marked as clock nets in the DB (run timeDesign first to build timing graph)

Command / Syntax	Description / Notes
report_clock_timing -type skew	Report clock skew
report_clock_timing -type latency	Report clock latency
set_interactive_constraint_modes [all_constraints]	Set all constraint modes active (for updating SDC)
set_timing_derate -delay_corner <corner> -data 10 -clock 5	Set timing derate for data/clock paths
reportTranViolation	Report transition (slew) violations
report_constraints -AllViolaters	Report all DRV (design rule) violations
add_ndr -width {Metal1 0.46 ...} -spacing {Metal1 0.3 ...} -name NDR2W2S	Add non-default routing rule (2x width, 2x spacing)
exportNdr NDR2W2S -lefFilename.lef	Export NDR rule to LEF file
setCTSMODE -routeLeafNonDefaultRule NDR2W2S	Apply NDR rule to leaf CTS nets
setCTSMODE -routeShielding VSS	Enable shielding of CTS nets with VSS
setCTSMODE -specMultiMode true	Enable multi-mode CTS specification
dbGet [dbGet -p1 top.nets.shieldNets.Name VSS].name	Get nets that are shielded with VSS
dbGet head.rules.name	Get all non-default rules loaded from LEF
dbGet [dbGet -p top.nets.isCTSclock 1].name	Get all CTS clock net names
dbGet top.nets.rule.name 2w2s	Get nets assigned to NDR rule '2w2s'
dbGet [dbGet -p top.nets.name netName].rule.name	Get NDR rule applied to a specific net
dbGet [dbGet -p1 top.insts.cell.terms.type clockTerm].name	Get all clock sink pin names
dbSet [dbGet -p top.nets.name netName].wires.status routed	Change routing status of a net (e.g. FIXED → ROUTED)

5. ROUTING COMMANDS

Command / Syntax	Description / Notes
route_design	Run global and detail routing (full flow command)
getAttribute -net net_name	Get attributes of a net
getNetWireLength net_name	Get total wire length of a net
describeCongestion	Print congestion summary for the design

Command / Syntax	Description / Notes
dumpCongestArea -all file_name	Dump all congested areas to a file
editAddRoute <X Y>	Begin interactive/ECO route at specified point
editCommitRoute <X Y>	Commit an interactive/ECO route
setEdit {specify metal and spacing}	Set edit mode for wire editing (layer/spacing)
editSelect -layer {metal1 metal2}	Select objects on specified routing layers
dbGet [dbGet -p top.nets.name netName].wires.layer.name	Get all routing layers used by a net
dbGet [dbGet -p2 top.insts.cell.isSequential 1].instTerms.cellTerm.type clockTerm].net.name	Get clock nets connected to sequential cell clock pins
dbGet [dbquery -area {x1 y1 x2 y2}].box	Query objects in a specified area
dbGet [dbGet -p head.LibCell.isInverter 1].name	Get all inverter library cell names
get_metric	Get all design QoR metrics (WNS, TNS, DRVs, etc.)
set_analysisismode -analysisType onChipVariation	Set OCV analysis mode for timing
all_fanout -from pin_name - endpoint_only	Get all fanout endpoints from a pin
report_clocks - uncertainty_table	Report clock uncertainty table
rcOut -spef file_name.spef - reg_corner corner_name	Write SPEF parasitics file for a given corner
createInstGroup group_name - guid box <coords>	Create an instance group with a guide box
CreateInstGroup group_Name - fence x1 y1 x2 y2	Create an instance group with a hard fence region
addInstToInstGroup group_name instName	Add an instance to an existing instance group
SetInstGroupPhyHier group_Name	Set physical hierarchy for an instance group
setDoAssign -buffer buf_name on	Enable buffer insertion during do-assignment step
ecoChangeCell -inst name - upsize	ECO: upsize a cell instance
setEcoMode -updateTiming false	Set ECO mode timing update on/off
EcoAddRepeater -cell DLY2X1 - net Net_Name -relativeDistToSink 0.1	Add a repeater near the sink (10% of distance to sink)
specifyNetWeight <net_name> <weight>	Specify relative routing weight/priority for a net
setCheckMode -Netlist true	Enable netlist checking mode
getCheckMode	Get current check mode settings

6a. Select Instance

Command / Syntax	Description / Notes
selectInst instName selectInst {inst1 inst2 ...}	Select one or more instances by name (wildcards allowed). <i>Highlights them in the design display window.</i>
deselectInst instName	Deselect a specific instance
deselectAll	Deselect all currently selected objects
SelectInst New1*	Select all instances whose name starts with 'New1'

6b. Select Net

Command / Syntax	Description / Notes
selectNet netName selectNet {net1 net2 net3}	Select and highlight net(s) in the design window. <i>Supports wildcards for net names.</i>
selectNet -clock	Select all clock nets
selectNet -nonDefaultRule	Select all nets with NDR (non-default routing rule)
selectNet -shield	Select all shielded nets
selectNet -allDefClock	Select nets with DEF USE CLOCK attribute
deselectNet netName	Deselect a specific net

6c. Select Pin

Command / Syntax	Description / Notes
selectPin pinName	Select an instance pin by name
selectIOPin pinName	Select an IO pin by name
selectPGPin -all selectPGPin -net netName selectPGPin -layer layerId selectPGPin -area llx lly urx ury	Select power/ground pins by all/net/layer/area filter
deselectPin pinName	Deselect a specific pin
deselectIOPin pinName	Deselect a specific IO pin

6d. Select Object (Generic)

Command / Syntax	Description / Notes
select_obj obj_ptr_list	Select objects using dbGet pointer list. Works for any object type (inst, wire, via, etc.).

Command / Syntax	Description / Notes
	<i>Example: select_obj [dbGet top.insts.name i1 -p]</i>
deselect_obj [dbGet top.insts.name i1 -p] deselect_obj -all	Deselect a specific object pointer or all objects
select_row -all select_row -name string select_row -site string	Select placement rows (all, by name, or by site)
selectGroup groupName	Select a floorplan group by name
deselectGroup groupName	Deselect a group by name
selectRouteBlk name selectRouteBlk -box x1 y1 x2 y2 name -layer 1 2	Select routing blockage by name and optionally by bounding box and layer
windowSelect x1 y1 x2 y2	Select all objects enclosed in the specified window area
windowDeselect x1 y1 x2 y2	Deselect all objects within the specified window area
windowToggleSelect x1 y1 x2 y2	Toggle selection of all objects in the specified area
lineSelect x1 y1 x2 y2	Select all objects crossed by the specified line
select_highlighted	Select all currently highlighted objects
selectModule DTMF_INST	Select a specific hierarchical module by name
deselectModule moduleName	Deselect a hierarchical module
reportSelect	Report current selection contents
dbGet selected	Return pointer to currently selected objects (Tcl)

6e. Select Instance By Cell Name

Command / Syntax	Description / Notes
selectInstByCellName cellName	Select all instances (including hierarchical) of a given cell type. <i>Child instances are also highlighted when an HInst is selected.</i>
deselectInstByCellName cellName	Deselect instances of a given cell type
selectInstOnNet netName	Select all instances connected to a specified net
deselectInstOnNet netName	Deselect instances on a specified net
dbGet [dbGet -p2 top.insts.cell.name cellName].name	dbGet query: list all instances of a specific cell type

7. HIGHLIGHT COMMANDS

Command / Syntax	Description / Notes
highlight -index 4	Highlight selected objects with Highlight Set #4 color/pattern

Command / Syntax	Description / Notes
highlight [dbGet -p1 top.insts.name instName] -index 3	Highlight a specific instance using index color
highlight [list [dbGet -p1 top.insts.name a] [dbGet -p1 top.insts.name b]] -index 4	Highlight a list of instances with Highlight Set #4
highlight -index 5 -color green -pattern dot4	Highlight with custom color and pattern (updates Highlight Set #5)
highlight -index 3 -color #33FF66 -pattern horizontal	Highlight using RGB hex color value with horizontal pattern
highlight -auto_color	Highlight selected objects with automatic (rotating) color
highlight -original	Highlight object in its original display color (useful with dim background)
highlight_pin -inst U1 -pin {E D} -color yellow -pattern solid -index 1	Highlight specific SIGNAL/POWER pins of an instance
highlight_pin_connection -from_pin ffa/Q -to_pin U7/A -mode flight_line	Highlight timing path segment between two pins
dehighlight	Remove highlighting from all highlighted objects
gui_dim_foreground -lightness_level dark	Dim the design foreground to highlight selected objects more clearly
highlightSyncObj	Sync highlight state of objects across views
set_object_color -object_name instName -color_id 3	Set a custom color for an instance or hierarchical module
set_object_color -object_name instName -color_id 3 -include_children color	Color an HInst and all its children with the same color
resetMultiColorsHier	Reset multi-color settings for hierarchical instances

8. GUI / VIEW COMMANDS

Command / Syntax	Description / Notes
win	Open the Innovus main GUI window
fit	Zoom/fit to show the full design in the window (bindkey: f)
zoomIn	Zoom into the design (bindkey: Z)
zoomOut	Zoom out from the design
zoomBox x1 y1 x2 y2	Zoom to a specific area defined by coordinates
zoomSelected [-margin value]	Zoom to fit the selected objects (default margin: 50%)
zoomTo x y zoomTo x y -radius 50	Zoom to a specific x,y coordinate with given radius (default: 150 microns)
viewLast	Return to the previous zoom/view (bindkey: w)

Command / Syntax	Description / Notes
viewNext	Go to the next saved zoom/view (bindkey: Y)
viewSnapshot -dir /path -name cts	View saved design snapshots (fplan, routing, etc.)
createSnapshot	Create a design snapshot with QoR parameters
gui_dump_picture filename.png gui_dump_picture filename - width 1000 -height 1000 gui_dump_picture -format GIF test1.gif	Save current layout as image (PNG/BMP/GIF/JPG/etc.)
dumpPictures	Save multiple layout pictures (placement, routing, congestion, etc.)
dumpToGIF filename.gif	Save a GIF snapshot of the current GUI view
pan panCenter panPage	Pan the design window
redraw	Redraw/refresh the design display
gui_deselect -all gui_deselect -rect {x1 y1 x2 y2} gui_deselect -point {x y}	Deselect all, by rectangle, or by point in GUI
gui_select -rect {x1 y1 x2 y2} gui_select -line {x1 y1 x2 y2} -append gui_select -rect {x1 y1 x2 y2} -toggle	Select objects by rect/line in GUI; use -append or -toggle
gui_attach_to_cursor {A B} gui_attach_to_cursor -selected	Attach objects to cursor for interactive placement
gui_group_hinst gui_ungroup_hinst	Group / ungroup hierarchical modules in GUI view
gui_open_cell_view gui_close_cell_view	Open / close the Cell Viewer
gui_get_schematic gui_schematic_add_selected gui_schematic_load_view gui_schematic_save_view	Schematic viewer commands: open, add selected, load/save view
gui_trace_flightline -inst instName -source -source_level 2 - source_color green	Trace flightlines from an instance for specified logic levels
gui_clear_trace_flightline	Clear all flightlines traced
gui_report_trace_flightline	Report traced flightline information
gui_save_flightline	Save current flightline view
gui_show_edge_number -objects instName gui_show_edge_number -objects * -reset	Show / clear edge numbers on partition/block objects
gui_dim_foreground - lightness_level {default medium dark}	Dim design foreground to different lightness levels (F12 bindkey)
add_gui_marker -name aa -color red -pt {100 100} -type X	Add a GUI marker at a specific location

Command / Syntax	Description / Notes
list_gui_marker	List all GUI markers placed
remove_gui_marker -name aa	Remove a specific GUI marker
add_gui_shape -rect {x1 y1 x2 y2} -layer layerName	Add an annotation shape to a GUI layer
delete_gui_object	Delete a GUI annotation object
db_browser obj_list head top selected	Launch DB Browser to inspect database objects
uiSetTool ruler uiSetTool select uiSetTool move uiSetTool obstruct	Set interactive GUI tool mode (ruler, select, move, blockage, etc.)
undo redo	Undo / redo the last GUI edit (flip, wire edit, move, etc.)
viewLog [-file logFileName]	Open the Innovus log file viewer
bindKey f "fit"	Bind a keyboard shortcut to a command
setPreference StretchRectilinear 1	Set GUI preference (stretch, snap, selection mode, etc.)
getPreference preference_name	Get current GUI preference value
getVersion	Get Innovus software version number

9. DESIGN FLOW & STATUS QUERIES

Command / Syntax	Description / Notes
dbGet top.statusIoPlaced	Check IO placement status
dbGet top.statusPlaced	Check standard cell placement status
dbGet top.statusClockSynthesized	Check CTS status
dbGet top.statusRouted	Check routing status
dbGet top.statusRCExtracted	Check RC extraction status
dbGet top.statusPowerAnalyzed	Check power analysis status
dbGet top.name	Get the top-level module name
dbGet top.terms.pins.allShapes.layer .name	Get metal layers of all IO pins
dbGet [dbGet -p top.insts.name instName].pgCellTerms.inOutDir bidij.name	Get PG pins with 'bidi' direction for a specific instance
foreach module_name [dbGet top.hInst.treeHInsts.cell.name] { puts "\$module_name" }	Print all hierarchical module names in the design
foreach leaf_name [dbGet -u	Print all unique leaf cell names used in the design

Command / Syntax	Description / Notes
<pre>top.insts.cell.name] { puts "\$leaf_name" }</pre>	
<pre>set netName <netName> set inst [dbGet [dbGet -p [dbGet -p top.nets.name \$netName].allTerms.isOutput 1].inst] puts "Driving inst: [dbGet \$inst.name] Cell: [dbGet \$inst.cell.name]"</pre>	Find driver instance name and cell name of a given net
<pre>dbGet [dbGet -p [dbGet -p2 top.terms.net.isClock 0].isInput 1].name</pre>	Find all non-clock input ports in the design
<pre>dbGet [dbGet -p top.insts.props {.name == "myProp" && .value == "xyzyz"}].name</pre>	Find instances with a specific user-defined property and value
<pre>set cellPtrList [dbGet -p head.allCells.name BUF*] foreach cellPtr \$cellPtrList { puts "[dbGet \$cellPtr.name] [dbFTermMaxCap ...]" }</pre>	Query max_cap for a set of library cells (BUF*)
<pre>dbGet [dbGet -p selected.cell.terms.name pinName].pins.allShapes.layer. extName</pre>	List all routing layers for a pin on the currently selected cell

10. TIMING COMMANDS

Command / Syntax	Description / Notes
timeDesign -reportsOnly -help	Get help on all timeDesign reporting options
report_timing [all_registers] -to [all_registers]	Report timing for all register-to-register paths
group_path -name r2r -from [all_reg] -to [all_reg]	Create path group for all reg-to-reg paths
report_timing -path_group r2r -max_paths 100 -nworst 10	Report top 10 worst paths in each of the 100 worst paths
report_timing -path_group r2r -max_paths 10 -nworst 10 -through <inst_name>	Report worst paths passing through a specific instance
report_timing -path_type full_clock	Report timing with full clock path detail
report_timing -path_type summary	Report timing in summary format
report_timing - machine_readable > file_name	Write machine-readable timing report to file
load_timing_debug_report filename	Load a machine-readable timing report for debug
report_timing -early	Report timing for hold (early/min) paths

Command / Syntax	Description / Notes
report_timing -collection	Return timing results as a collection object
set x [report_timing -from [all_reg] -to [all_reg] -collection] get_property \$x slack get_property \$x path	Query properties from a timing collection (slack, path, etc.)
get_property [all_clocks] source get_property [all_clocks] period get_property [all_clocks] waveform	Get clock source, period, and waveform
set_propagated_clock	Set clocks to propagated (non-ideal) mode
clearClockDomains setAnalysisMode -honorClockDomains false	Clear clock domains and disable domain-aware analysis
group_path -name to_srams -to top/inst_sram*/D* setPathGroupOptions to_srams -slackAdjustment 1000 -slackAdjustmentPriority 10	Create and configure a path group to SRAM D-inputs
analyze_paths_by_drv -generate_tran_file <file> -generate_cap_file <file> -generate_fanout_file <file>	Generate DRV analysis files for transition, capacitance, fanout
set_analysis_view -setup view_name -hold view_name -update_timing	Set active timing analysis view for setup/hold
all_constraint_modes all_constraint_modes -active	List all / active constraint modes
report_timing -net -from startPoint -to endpoint -path_type full_clock -through instName	Detailed timing path report with net delays and through-instance
get_property [get_property [get_clocks] clock_network_pins] actual_latency_late_fall_max	Get actual clock latency (late fall max) at clock network pins

11. SCAN CHAIN COMMANDS

Command / Syntax	Description / Notes
specifyScanChain chain_name <name> -start <i/p pin> -stop <o/p pin>	Define a scan chain with start/stop pins
scanTrace	Report scan chain start/end points and elements
scanReorder	Reorder scan chain elements for minimum wire length
displayScanChain	Display scan chain connectivity in the GUI

REFERENCE NOTES

dbGet Flags: -p (filter returning parent pointer), -p1 (filter returning 1st level up pointer), -p2 (filter returning 2nd level up pointer), -u (unique objects only), -regexp (use regexp matching)

dbSet: Used to modify database properties. Always pair with dbGet to get the object pointer first. Example: dbSet [dbGet -p top.nets.name netName].wires.status routed

Wildcards: The * wildcard is supported in most name-matching parameters (instName, cellName, netName, etc.). Example: selectInst reg* selects all instances starting with 'reg'.

Source Documents: [Cadence Innovus Text Command Reference v23.11 \(5163 pages\)](#) | [dbGet Commands Guide](#) | Internal PD team command notes